

Inter-Animal Contact Geometry from Segmentation Masks

Do mask-derived contact features beat keypoint-hull geometry
for mouse social-behaviour recognition?

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Abstract

Automated recognition of *social* animal behaviour — chasing, huddling, investigating — hinges on the geometry of how two bodies meet. The field reads that geometry from *keypoint hulls*: convex polygons over a handful of pose keypoints, used by tools such as SimBA [2] and MARS [3]. But a hull over sparse keypoints is crudest *exactly during contact*, when keypoints occlude, swap, or hallucinate and the polygon smooths over the real boundary where two silhouettes touch. This proposal asks a sharp, testable question: does *inter-animal contact geometry derived from segmentation masks* (mask IoU/overlap, contact-boundary length) add signal over keypoint-hull geometry for social-behaviour recognition? We test it on the MABe22 mouse-triplet benchmark [1] with SAM2 [4] as the enabling segmentation technology, every heavy model frozen, and a single small head trained on cached features. Shape-from-silhouette is *not* claimed novel — it is established in human action and gait recognition [6, 7, 8]; the contribution is, to our knowledge, the first use of **mask-contact geometry** as an explicit behavioural feature, in a clean head-to-head against the hull geometry the field already uses. The answer is interpretable either direction.

1 Background and motivation

Recognising what animals are *doing* from video underpins neuroscience and ecology, and *social* behaviours — where two or more animals interact — are both the most scientifically interesting and the hardest to read. The defining signal of a social behaviour is **geometric**: how close two bodies are, how much they overlap, where they touch.

The standard way to compute that geometry is from **pose keypoints**. Tools such as SimBA [2] build a convex *hull* (polygon) over each animal’s keypoints and derive features like polygon percentage overlap and difference-area; MARS [3] similarly derives relational features from tracked keypoints for mouse social behaviour. This works, but it inherits the weakness of sparse pose *precisely when it matters most*. During close contact — mounting, huddling, anogenital investigation — keypoints occlude one another, the tracker swaps identities between near-identical animals, and a hull over a few noisy points bears little relation to the true contacting surface. The crudeness of the representation peaks exactly at the moment the behaviour is defined.

A **segmentation mask** does not have this failure mode: it captures the animal’s full 2D extent and, between two masks, the actual boundary where the silhouettes meet. Silhouette-based shape is itself an old, well-validated idea — motion-energy/history templates with Hu moments [6], “actions as space-time shapes” [7], and the entire field of gait recognition [8]. What changed is the *extraction*: the classical silhouette came from background subtraction against a static clean background, which multi-animal lab/field video denies. **SAM2** [4] — promptable, video-propagated instance segmentation — recovers a tracked foreground silhouette where the

classical mechanism fails. That makes mask-based contact geometry newly *computable*, and nobody has yet used it as a behavioural feature.

Two adjacent results sharpen the case. On the MammAlps wildlife benchmark, adding a static *scene/background* segmentation map did not merely fail to help — it *degraded* recognition (video-only class-average mAP fell from 0.453 to 0.403). PanAf-FGBG [5] showed wildlife models over-rely on behaviour-correlated background and generalise poorly across cameras, using SAM 2 masks to *erase* the animal and study the background. Both motivate the *opposite* move: keep the foreground geometry and discard the background.

2 Research questions and hypotheses

RQ3 (primary)

Does *mask-derived inter-animal contact geometry* (mask IoU/overlap, contact-boundary length) improve recognition of social/contact behaviours over the *keypoint-hull geometry* the field uses?

RQ4 (insurance)

Is a shape/geometry representation more robust under *background shift* (unseen cameras) than background-contaminated RGB features — the inversion of PanAf-FGBG? Publishable even if in-distribution accuracy is flat.

RQ1 (redundancy)

Does explicit mask geometry add anything *over a frozen VideoMAE*, which already “sees” the silhouette in pixels? (Confronted directly in the fusion stage.)

Hypothesis. Mask-contact geometry carries contact information that keypoint-hull geometry captures only crudely, with the gap largest on the deepest-contact behaviours. Fusing it will (i) beat the hull baseline head-to-head on contact behaviours, (ii) show its clearest per-behaviour gains where keypoints occlude/swap, and (iii) degrade more gracefully than RGB under background shift.

3 Proposed approach

Dataset. *MABe22 mouse-triplet* [1], the labelled *submission* split (the one that ships video, keypoints *and* labels; verified by inspection): **1,830 one-minute sequences** at 30 Hz (1,800 frames each; 224×224 grayscale top-view), three near-identical mice, and **dense framewise binary labels** for four social behaviours — **chases**, **huddles**, **oral_oral_contact** (oral contact), **oral_genital_contact**. The behaviours are heavily imbalanced (huddle $\approx 8\%$, chase $\approx 0.05\%$ of frames positive), so we report **average precision** alongside F1 and class-balance at train time. Multi-animal contact is native here, which single-animal wildlife clips (MammAlps) cannot provide; MammAlps is retained as a secondary single-animal / background-robustness setting.

Keypoints — given, not estimated. The dataset ships per-frame tracked pose as an array $K \in \mathbb{R}^{T \times 3 \times 12 \times 2}$ (1,800 frames $\times$ 3 mice $\times$ 12 keypoints $\times$ x, y). We run *no* pose estimation. Crucially, **both geometry conditions start from these same keypoints**, so the comparison isolates the *representation* (sparse polygon vs. full silhouette), not the pose source.

Masks — SAM2 prompted by the keypoints. For each mouse i at frame t we turn its 12 keypoints into SAM2 prompts — the keypoints as positive point prompts plus a tight box from their extent — run SAM2 once, and *propagate* the mask across the clip for temporal and identity consistency, yielding one binary mask M_i^t per mouse per frame. SAM2 is inference-only (Table 1); masks (or the geometry derived from them) are cached once. Whether the three masks stay *separate and identity-stable through the contact frames* is the single make-or-break risk, settled by the week-1 pilot (§6).

The two geometry conditions (identical downstream). From the same per-frame inputs we compute, for each of the three mouse-pairs (a, b) , a small contact-feature vector — once from *hulls* (baseline) and once from *masks* (novel):

- **Hull contact geometry (baseline — SimBA/MARS).** Take each mouse’s convex hull $H_i^t = \text{conv}(K_i^t)$, a polygon over its 12 keypoints. Per pair: overlap ratio $O_{ab} = |H_a \cap H_b| / |H_a \cup H_b|$ (SimBA `polygon_pct_overlap`), difference-area, centroid distance, and minimum inter-keypoint distance. Polygon algebra via `shapely`; essentially free.
- **Mask contact geometry (novel).** From the two binary masks, the silhouette overlap $\text{IoU}_{ab} = |M_a \cap M_b| / |M_a \cup M_b|$ and the *contact-boundary length* $L_{ab} = |\partial M_a \cap (M_b \oplus B_r)|$ — the count of mouse-*a* boundary pixels lying within r px of mouse-*b*’s mask (∂ = boundary, $\oplus B_r$ = dilation by radius r), i.e. *how much of the two silhouettes physically touch*. Plus per-mouse **shape descriptors** for the posture side: area, aspect ratio, solidity (area / convex-hull area), and the seven Hu moments.

Each condition yields a per-frame vector (3 pairs $\times$ pair-features, + 3 mice $\times$ shape-features) fed to a **tiny temporal encoder + small head / linear probe**. Head capacity is held *identical*; the *only* thing that changes between baseline and novel condition is whether pair-overlap comes from a hull polygon or a mask — so any gain reflects information, not parameters. The mechanistic bet: O_{ab} (hull) is crudest *exactly* during deep contact, where keypoints occlude/swap and the polygon smooths over the true boundary, whereas IoU_{ab} and L_{ab} (mask) read the actual contacting surface.

Component	Pretrained model (frozen)	Role
Masks	SAM2 [4]	frame + keypoint prompt $\rightarrow$ per-mouse <i>mask</i>
Appearance	VideoMAE (Kinetics)	clip $\rightarrow$ embedding (fusion / redundancy test)
Geometry	— (computed from masks/keypoints)	mask IoU, contact-boundary; hull overlap (baseline)

Table 1: Frozen building blocks. SAM2 is inference-only (never trained); the geometry features are deterministic; only a small head / linear probe is trained. The *keypoint-hull* geometry (SimBA/MARS-style) is the comparison condition and is essentially free from the given keypoints.

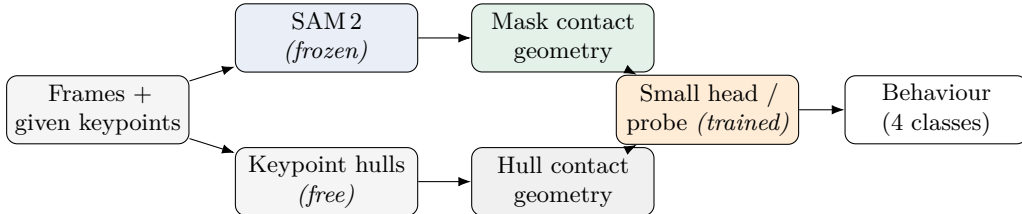


Figure 1: The head-to-head: identical-capacity head on *mask* contact geometry vs. *hull* contact geometry (the field’s baseline), from the same keypoints. Heavy models frozen and cached; the mask branch is the novelty.

Training & protocol. Only the head trains, on cached features (M4 / PyTorch MPS). Class-balanced sampling; fixed seeds; report **mean $\pm$ 95 % CI over ≥ 5 seeds**. Modality dropout [9] lets the geometry stream be dropped at test time; a heavier DINOv2 [10] masked-crop embedding is held in reserve only if hand-crafted features underperform.

4 Novelty (stated honestly)

The contribution is *not* a new shape representation, and not “fuse masks and tabulate it.” It is, to our knowledge, the **first use of inter-animal segmentation-mask contact geometry (mask IoU, contact-boundary length) as explicit features for social/contact behaviour recognition**, in a controlled comparison against the keypoint-hull geometry the field already uses. Each near neighbour is distinct:

- **SimBA [2] / MARS [3]**: two-body geometry from *keypoint hulls*, not masks. This is exactly our baseline condition; the experiment is mask-overlap vs hull-overlap, head to head.
- **Silhouette/shape methods [6, 7, 8]**: established in *human* action and gait recognition, via background-subtraction silhouettes (single subject, clean background). We adapt the idea to *multi-animal contact* using SAM2 where background subtraction is impossible — and gait is identity, not behaviour.
- **PanAf-FGBG [5]**: same tool (SAM2) for the *opposite* purpose — masks used to *erase* the animal and study the background. We keep the foreground geometry as a positive signal, and use their robustness finding as motivation.

5 Evaluation and pre-registered ablations

Eval contract (locked before modelling). Framewise prediction of the four behaviours (treated as independent binary tasks), over the annotated clips. **Leakage-safe split by sequence** — every frame of a sequence stays in one split; never a random frame/window. Metric: per-behaviour **F1** (the MABe convention) *and* average precision (for comparability with the MammAlps secondary track), macro-averaged over the four behaviours. The metric/split are model-agnostic; the *model* (a linear probe for the clean feature comparison, or the small fusion head) is separate.

Pre-registered ablations. To keep a small gain interpretable against seed noise, the conditions and the success criterion are fixed *in advance*: (a) **hull geometry alone** (the baseline); (b) **mask geometry alone** vs (a) — the key isolation; (c) **fusion** (mask geometry on top of frozen VideoMAE) — the redundancy test (RQ1). Each runs ≥ 5 seeds, identical head capacity, reported as mean $\pm$ 95 % CI plus per-behaviour scores. **Success is decided up front:** mask geometry helps iff it beats hull geometry with a 95 % CI excluding zero *and* a pre-registered minimum effect, or a clear per-behaviour win on the deepest-contact behaviours (huddle, anogenital sniff). A scrambled-mask control rules out capacity effects.

6 Risks and contingency

The novelty and the technical risk are the same thing: **can SAM2 keep three near-identical mice as separate, identity-stable instances through the contact frames**, when masks tend to merge or swap? A **week-1 masking pilot** on the deepest-contact clips (huddle, chase) settles it, with a written decision rule:

- **PASS** (identities hold through contact) $\rightarrow$ proceed; contact geometry is the headline.
- **PARTIAL** (masks merge only in deepest contact) $\rightarrow$ treat a merge event as a coarse *high-contact* feature alongside IoU/boundary on clean frames; proceed and say so.
- **FAIL** (frequent swaps/merges) $\rightarrow$ pivot the headline to **background-robustness (RQ4)** on the secondary MammAlps track, and report the contact-geometry attempt as an exploratory/negative result with the mask-quality analysis as evidence. The dissertation keeps a clean spine either way.

Compute is bounded: SAM2 is run once, behaviour-windowed and frame-subsampled ($\sim 5\text{--}6$ Hz), caching derived geometry rather than raw masks; the trained head sees only cached vectors.

Indicative timeline

- Wk 1** Acquire MABe22; lock the label/eval contract; **masking pilot + decision**.
- Wk 2** Scoped SAM2 mask cache (Kaggle); begin geometry features.
- Wk 3** Finish contact-geometry + hull baseline extractor; head overfit sanity.
- Wk 4** **Mask-geometry vs hull-geometry head-to-head** — first real result.
- Wk 5** Per-behaviour analysis; multi-seed CIs; go/no-go on the MammAlps secondary track.
- Wk 6** Fusion / VideoMAE-redundancy test; lock primary; begin secondary if reached.
- Wk 7** Background-shift / robustness (RQ4); consolidate analysis.
- Wk 8** Write-up; figures; package cached-feature pipeline + code.

References

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